ECON104 Midterm Review

Winter 2025

Logistics

- Exam is Thursday in class from 12:30 - 1:40PM.
- Topics are heteroskedasticity, time series, endogeneity, and simultaneous equations.
- Bring your student IDs and a calculator

What to study

- Practice midterm is the best study tool for the midterm; style of question for multiple choice questions are similar.
- For the free response questions, think about how you would answer the practice questions *without* the given answer choices.
    - Discussion problems are also useful to study, especially the "theory" question(s) that are often the first question of the selected questions. Note that the coding discussion problems are not as relevant and are generally more difficult because you won't need to do "live" coding.
- There will still be problems that require you to interpret code, and not just code output
    - Labs (including that Econ 103 lab PDF) are the best tools to study the necessary code.
    - Reading/understanding code output is still required!

Other Code Tips

When studying the lab codes, think about the main takeaways/lessons of the labs. There is more to keep in mind, but think about the following questions:

- Lab 1: How do I change standard errors from OLS to HC standard errors? How do I implement WLS? How do I (manually) conduct a GQ or BP type test?
- Lab 2: What is the difference between setting up a time series regression vs. a standard OLS model? How do I know when there is or is not autocorrelation?
- Lab 3: What is the difference between the lm and ivreg commands? How do I know what the instruments, endogenous, and exogenous variables are by reading a command? How do I interpret the diagnostic tests in the ivreg output?
- Lab 4: How do you apply the concepts of two stage least squares to a simultaneous equation system?

Heteroskedasticity Overview

Given data (x_i, y_i), we may estimate a linear model:

$$y_i = \beta_1 + \beta_2 x_i + e_i \tag{1}$$

Heteroskedasticity Overview

- How do we test for it?
  1. Breusch-Pagan (BP) Test
     - White Test
  2. Goldfield-Quandt (GQ) Test
- How do we fix it?
  1. White Standard Errors (HC)
  2. Weighted Least Squares

Breusch-Pagan (BP) Test

After estimating the original model, retrieve the estimated residuals $\hat{e}_i$. The BP test uses the following linear model:

$$\hat{e}_i^2 = \theta_0 + \theta_1 z_1 + \cdots + \theta_J z_J + v_i \tag{2}$$

where $(z_1, \ldots, z_J)$ are the set of J variables that you believe explain the heteroskedasticity in your model. The test statistic is then given by:

$$\hat{BP} = N \cdot R^2 \sim \chi^2_{(J)} \tag{3}$$

If $\hat{BP} \geq \chi^2_{(1-\alpha,J)}$), then the model is heteroskedastic

White Test

The White Test is a special version of the BP test where you use the full set of first and second order terms of the X variables from your original equation. For example, if you have the original model

$$Y_i = \beta_1 + \beta_2 X_1 + \beta_3 X_2 + e_i \tag{4}$$

the White Test becomes

$$\hat{e}_i^2 = \theta_0 + \theta_1 X_1 + \theta_2 X_2 + \theta_3 X_1^2 + \theta_4 X_2^2 + \theta_5 X_1 X_2 + v_i \tag{5}$$

Goldfield-Quandt (GQ) Test

The GQ test is a test on **GROUPS**; We want to see if $\sigma_1^2 = \sigma_2^2$

1. Arrange the observations in ascending order of the variable you believe is the source of heteroskedasticity (Skip if discrete)
2. Split into two groups
3. Run two separate regressions and compute their variance:

$$\hat{\sigma}_g^2 = \frac{SSE_g}{N_g - K_g}$$

4. Compute the test statistic (with $\sigma_1 > \sigma_2$):

$$\hat{GQ} = \frac{\hat{\sigma}_1^2}{\hat{\sigma}_2^2} \sim F(N_1 - K_1, N_2 - K_2)$$

5. If $\hat{GQ} > F_\alpha(N_1 - K_1, N_2 - K_2)$, then heteroskedasticity is present

White's HC Standard Errors

Recall the original OLS standard errors:

$$\text{Var}(b_2|X) = \frac{N}{N-K}\frac{\sum_{i=1}^{N}[(x_i-\bar{x})^2\hat{\sigma}^2]}{[\sum_{i=1}^{n}(x_i-\bar{x})^2]^2} \tag{6}$$

One solution to heteroskedasticity is computing different standard errors. One example is White's HC1 standard error:

$$\text{Var}(b_2|X) = \frac{N}{N-K}\frac{\sum_{i=1}^{N}[(x_i-\bar{x})^2\hat{e}_i^2]}{[\sum_{i=1}^{n}(x_i-\bar{x})^2]^2} \tag{7}$$

The only difference is we substitute in the squared residuals ($\hat{e}_i^2$) for the error variance ($\hat{\sigma}^2$) in the numerator!

Weighted Least Squares

Suppose heteroskedasticity is present and we believe that it takes on the following form:

$$\text{Var}(e_i|x_i) = \sigma^2 h(x_i) \tag{8}$$

If $h(x_i)$ is known, then we can "undo" the heteroskedasticity by weighting the regression by $\sqrt{h(x_i)}$:

$$\begin{aligned} \frac{y_i}{\sqrt{h(x_i)}} &= \beta_1 \frac{1}{\sqrt{h(x_i)}} + \beta_2 \frac{x_i}{\sqrt{h(x_i)}} + \frac{e_i}{\sqrt{h(x_i)}} \\ y_i^* &= \beta_1 x_{0i}^* + \beta_2 x_{1i}^* + e_i^* \end{aligned} \tag{WLS}$$

Weighted Least Squares

What affect does this have on the variance of our residuals?

$$
\begin{aligned}
\mathrm{Var}(e_i^*|x_i) &= \mathrm{Var}\left(\frac{e_i}{\sqrt{h(x_i)}}\middle| x_i\right) \\
&= \frac{1}{h(x_i)}\mathrm{Var}(e_i|x_i) = \frac{1}{h(x_i)}\sigma^2 h(x_i) \\
&= \sigma^2
\end{aligned}
$$

After our weighting, we have a homoskedastic model!

→ Our parameters β_1 and β_2 are unchanged

→ The standard errors of the model are changed:

$$\mathrm{Var}(b_2|X)_{WLS} = \frac{N}{N-K}\frac{\sum_{i=1}^{N}[(x_i^* - \bar{x}^*)^2\hat{\sigma}^2]}{\left[\sum_{i=1}^{n}(x_i^* - \bar{x}^*)^2\right]^2} \tag{9}$$

Generalized Least Squares

In WLS, we assumed that $h(x)$ followed a certain function, but what if we don't know what it is?
$\rightarrow$ We need to get an estimate of $h(x)$!

(1) Suppose J variables are in your skedastic function and assume it takes the form:

$$h(x) = \prod_{j=1}^{J} x_{ij}^{\theta_j} \tag{10}$$

(2) As $\mathbb{E}[e_i] = 0$ by assumption, we create the model

$$\text{Var}(e_i|x_i) = \mathbb{E}[e^2] = \sigma^2 \prod_{j=1}^{J} x_{ij}^{\theta_j} \tag{11}$$

(3) Taking logs of both sides yields

$$\log(\mathbb{E}[e_i^2]) = \theta_0 + \theta_1 \log(x_{i1}) + \cdots + \theta_j \log(x_{iJ}) \tag{12}$$

Generalized Least Squares

(4) Using sample residuals an an error term gives us a linear model!

$$\log(\hat{e}_i^2) = \theta_0 + \theta_1 \log(x_{i1}) + \cdots + \theta_j \log(x_{iJ}) + v_i \tag{13}$$

(5) Run OLS on this model to receive parameter estimates for $\theta_0, \theta_1, \ldots, \theta_J$ and compute weights:

$$\hat{w}_i = e^{\hat{\theta}_0} \prod_{j=1}^{J} x_{ij}^{\hat{\theta}_j}$$

(6) Run GLS on your original model with weights $\sqrt{w_i}$:

$$\frac{y_i}{\sqrt{w_i}} = \beta_1 \frac{1}{\sqrt{w_i}} + \beta_2 \frac{x_i}{\sqrt{w_i}} + \frac{e_i}{\sqrt{w_i}} \tag{14}$$

WLS vs GLS

- WLS assumes you know the function $h(x)$, GLS estimates this function:
  - WLS might have specification error
  - GLS might have estimation error
  - With error, some heteroskedasticity may remain.
    $\longrightarrow$ Add on a White standard error and you are good!
- The residual variances for the two methods differ:
  - WLS has $\text{Var}(\hat{e}_i^2) = \sigma^2$
  - GLS has $\text{Var}(\hat{e}_i^2) = 1$
  - This happens because the intercept term in the GLS function θ_0 corresponds to σ^2:

$$\hat{w}_i = \hat{\sigma}^2 \hat{h}(x_i)$$

Time Series

Suppose that instead of observing a cross-section of units in a single period, we observe a single individual over several time periods? In particular, we now have data $(X_t, Y_t)_{t=1}^{T}$ where the observations are indexed across time periods.

In these settings, oftentimes we assume that there is some relationship between past observations and future outcomes. The idea of time series econometrics is building models to estimate those relationships.

Stationarity

A time series is said to be **stationary** if its distribution does not change over time:

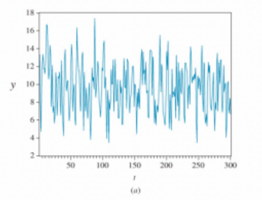


图 1: Stationary variable

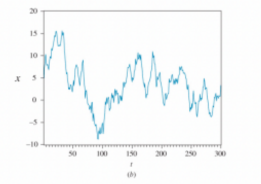


图 2: "Wandering" Non-stationary

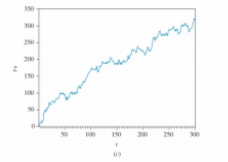


图 3: Trending non-stationary

3 Types of Time Series Models

- **Autoregressive:** Relates previous values of Y with current outcomes of Y:

$$Y_t = \beta + \beta_1 Y_{t-1} + \cdots + \beta_p Y_{t-p} + e_t \qquad \text{(AR(p))}$$

- **Distributed Lag:** Relates previous values of X with current outcomes of Y:

$$Y_t = \delta + \delta_0 X_t + \delta_1 X_{t-1} + \cdots + \delta_q X_{t-q} + e_t \qquad \text{(DL(q))}$$

- **ARDL:** Combines the previous two models into one:

$$Y_t = \theta + \beta_1 Y_{t-1} + \cdots + \beta_p Y_{t-p} + \delta_0 X_t + \delta_1 X_{t-1} + \cdots + \delta_q X_{t-q} + e_t \qquad \text{(ARDL(p,q))}$$

Autocorrelation

When observing a variable W across multiple time periods, we can compute its correlation with itself (**autocorrelation**):

$$\rho_s = \frac{\text{Cov}(W_t, W_{t-s})}{\text{Var}(W_t)} \tag{15}$$

In our time series model, we want to make sure that our error term does not have autocorrelation:

$$\rho_s = \frac{\text{Cov}(e_t, e_{t-s})}{\text{Var}\, e_t} = 0 \text{ for all lags } s$$

This is analogous to our independence of errors assumption from OLS

Testing Autocorrelations

How can we tell whether the autocorrelation is significantly different from zero?

Our test statistic is

$$Z = \sqrt{T} \cdot \hat{\rho}_s \sim N(0,1) \tag{16}$$

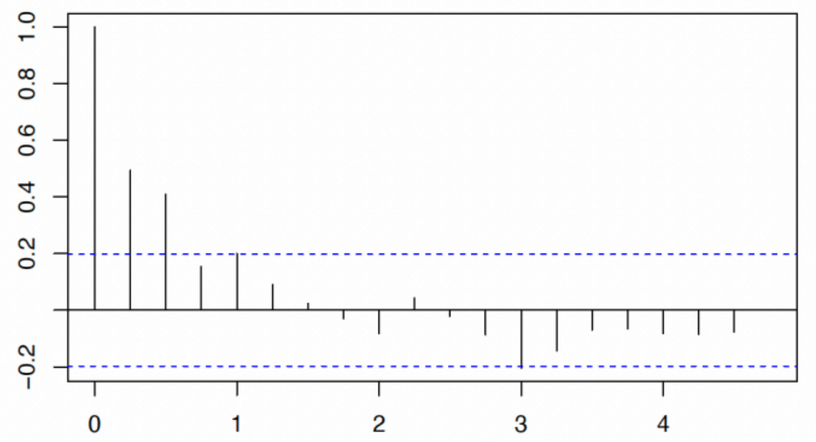
Our test hypotheses are:

$$H_0 : \rho_s = 0 \quad \text{(No autocorrelation)}$$
$$H_1 : \rho_s \neq 0 \quad \text{(Autocorrelation)}$$

If we reject the null, then there is autocorrelation of order s.

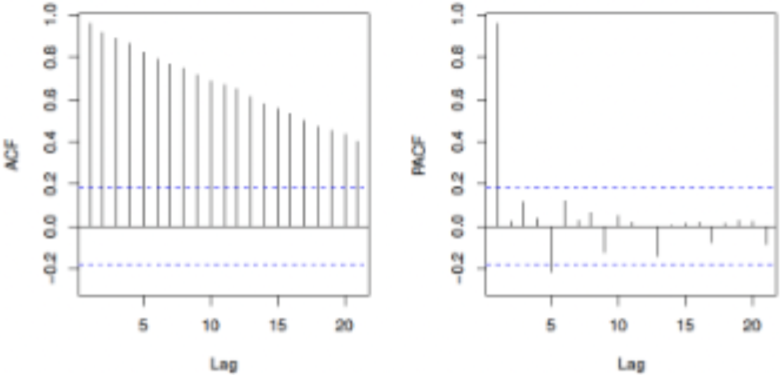
Correlograms

The correlogram is a representation of sample correlations at various lengths. A horizonal line is drawn at $\pm 2/\sqrt{T}$, which is approximately the critical value:



Partial Autocorrelations

Idea is to "partial out" the intermediate lags of the ACF to isolate the lag of interest.



Breusch-Godfrey (BG) Test

A Breusch-Godfrey test is a regression of today's errors on past errors. Suppose you had the following ARDL model:

$$y_t = \delta + \theta_1 y_{t-1} + \delta_1 x_{t-1} + e_t \tag{17}$$

and wanted to know if the errors had autocorrelation at some lag s?

Solution: Estimate the equation, get the residuals $\hat{e}_t$, and run the following regression:

$$\hat{e}_t = \delta + \rho_1 \hat{e}_{t-1} + \cdots + \rho_s \hat{e}_{t-s} + v_t \tag{18}$$

Perform an $LM = T \times R^2$ test for autocorrelation with the test statistic from the $\chi^2_{(s)}$ distribution. (Rejection = autocorrelation)

Model Selection

- Choosing the order, p (Lags of Y)
  1. Check correlogram of errors after estimating an AR(1) model, include the significant lags as new Y variables. Repeat with new correlogram.
  2. Compute a PACF correlogram of the Y variables and select significant lags
- Choosing the order, q (Lags of X)
  1. Begin by selecting the maximum length you are willing to consider.
  2. Create alternate models of different lag lengths.
  3. Using some criterion (AIC, BIC/SC), pick the model with the lowest value

HAC Standard Errors

If you can't fully eliminate autocorrelated error terms, then you'll need to compute standard errors that are robust to it. These are called HAC errors.

The most commonly used one is the Newey-West Standard Error:

$$\hat{SE}_{NW} \approx \hat{SE}_{HC1}(1+g) \tag{19}$$

where $\hat{SE}_{HC1}$ are the White standard errors from heteroskedasticity, and g is some function of the autocorrelations.
In general, putting in HAC standard will lead to wider confidence intervals.

AR(1) Error to ARDL Model

Suppose we write the following model

$$y_t = \alpha + \beta_0 x_t + e_t \tag{20}$$

but we believe the error terms follow the following process:

$$e_t = \rho e_{t-1} + v_t \tag{21}$$

How would we correctly write the model to get rid of autocorrelation in the errors?

AR(1) Error to ARDL Model

1. Plug in the error formula into your original equation:

$$y_t = \alpha + \beta_0 x_t + \rho e_{t-1} + v_t$$

2. Rewrite the error term for $t-1$ with the original model

$$e_{t-1} = y_{t-1} - \alpha - \beta_0 x_{t-1}$$

3. Plug into the model and rearrange terms:

$$\begin{aligned} y_t &= \alpha + \beta_0 x_t + \rho(y_{t-1} - \alpha - \beta_0 x_{t-1}) + v_t \\ y_t &= \delta + \theta y_{t-1} + \beta_0 x_t + \beta_1 x_{t-1} + v_t \end{aligned}$$

This is now an ARDL(1,1) model where the parameters have some relation to each other.

ARDL(1,1) to IDL

Suppose we had an ARDL(1,1) Model:

$$y_t = \delta + \theta_1 y_{t-1} + \delta_0 x_t + \delta_1 x_{t-1} + e_t \tag{22}$$

and wanted to translate it to an Inifnite Distributed Lag (IDL) model of the following form:

$$y_t = \alpha + \beta_0 x_t + \beta_1 x_{t-1} + \beta_2 x_{t-2} + \cdots + v_t \tag{23}$$

How would we do it?

ARDL(1,1) to IDL

1. Write the IDL model for y_{t-1}:

$$y_{t-1} = \alpha + \beta_0 x_{t-1} + \beta_1 x_{t-2} + \beta_2 x_{t-3} + \cdots + v_{t-1}$$

2. Plug it into your original ARDL model:

$$\begin{aligned} y_t = \delta + \theta_1(\alpha + \beta_0 x_{t-1} + \beta_1 x_{t-2} + \beta_2 x_{t-3} + \cdots + v_{t-1}) \\ + \delta_0 x_t + \delta_1 x_{t-1} + e_t \end{aligned}$$

3. Match up terms according to the lags of the x variables:

$$y_t = (\underbrace{\delta + \theta_1 \alpha}_{\alpha}) + \underbrace{\delta_0}_{\beta_0} x_t + (\underbrace{\delta_1 + \theta_1 \beta_0}_{\beta_1}) x_{t-1} + \underbrace{\theta_1 \beta_1}_{\beta_2} x_{t-2} + \cdots + u_t$$

This process generalizes to different orders of the ARDL model.

Total Effects in IDL Models

If we are interested in knowing what a one unit change in x has on future outcomes of y, we would compute the sum:

$$TE = \sum_{k=0}^{\infty} \beta_k$$

Assuming $\beta_k = \beta_0 r^k$ where $|r| < 1$, this infinite sum is given by the formula:

$$TE = \sum_{k=0}^{\infty} \beta_0 r^k = \frac{\beta_0}{1 - r}$$

This is known as a Geometric Series!

Forecasting

Suppose we have an AR(1) model and we wanted to make a forecast, or prediction, for the next three periods. Assuming our model is correct, what would the next three periods true outcomes be?

$$y_{t+1} = \theta_0 + \theta_1 y_t + e_{t+1} \tag{24}$$
$$y_{t+2} = \theta_0 + \theta_1 y_{t+1} + e_{t+2} \tag{25}$$
$$y_{t+3} = \theta_0 + \theta_1 y_{t+2} + e_{t+3} \tag{26}$$

But what would our forecasts be?

Forecasting

In general, our forecasts will be of the form:

$$\hat{y}_{t+s} = \mathbb{E}[y_{t+s}|\mathcal{I}_t] \tag{27}$$

where $\mathcal{I}_t$ is all the data you have today. In our example, that creates forecasts

$$\hat{y}_{t+1} = \hat{\theta}_0 + \hat{\theta}_1 y_t \tag{28}$$

$$\hat{y}_{t+2} = \hat{\theta}_0 + \hat{\theta}_1 \hat{y}_{t+1} \tag{29}$$

$$\hat{y}_{t+3} = \hat{\theta}_0 + \hat{\theta}_1 \hat{y}_{t+2} \tag{30}$$

While creating point forecasts might be straightforward, how would we compute the variance of our forecast errors

$$f_s = y_{t+s} - \hat{y}_{t+s}?$$

Forecast Errors

Given the definition of forecast errors, we can begin computing them for our current model:

$$f_1 = y_{t+s} - \hat{y}_{t+s} = (\theta_0 - \hat{\theta}_0) + (\theta_1 - \hat{\theta}_1)y_t + e_{t+1}$$

One simplifying assumption that we use is $\hat{\theta} = \theta$. This assumption is reasonable as in large samples we have $\hat{\theta} \approx \theta$ (unbiased), and is useful as it gets rid of a lot of terms. The forecast errors can then written as:

$$\begin{aligned}
f_1 &= e_{t+1} \\
f_2 &= \theta_1(y_{t+1} - \hat{y}_{t+}) + e_{t+2} = \theta_1 f_1 + e_{t+2} \\
f_3 &= \theta_1 f_2 + e_{t+3}
\end{aligned}$$

Forecast Errors

Forecast errors can then be derived as the following:

$$
\begin{aligned}
\text{Var}(f_1) &= \text{Var}(e_{t+1}) = \sigma^2 \\
\text{Var}(f_2) &= \text{Var}(\theta_1 f_1 + e_{t+2}) = (\theta_1^2 + 1)\sigma^2 \\
\text{Var}(f_3) &= \text{Var}(\theta_1 f_2 + e_{t+3}) = (\theta_1^4 + \theta_1^2 + 1)\sigma^2
\end{aligned}
$$

We can then construct confidence intervals for our forecasts with:

$$CI_s = \hat{y}_{t+1} \pm t_c \cdot \sqrt{\text{Var}(f_s)} \tag{31}$$

Granger Causality

Suppose we had an ARDL(p,q) model

$$y_t = \delta + \sum_{s=1}^{p} \theta_s y_{t-s} + \sum_{k=0}^{q} \delta_k x_{t-k} + e_t \tag{32}$$

and we wanted to know if the lags of X were impact the outcome of Y. A Granger Causality Test is a F test that tests the joint signiciance of the δ_k terms:

$$\begin{aligned} H_0 :& \delta_0 = \delta_1 = \cdots = \delta_q = 0 \\ H_1 :& \text{At least one is not equal to } 0 \end{aligned}$$

Exogeneity

Given data (x_i, y_i), we may estimate a linear model:

$$y_i = \beta_1 + \beta_2 x_i + e_i \tag{33}$$

Originally, we assumed that the error was uncorrelated with our x variable, i.e our x variable is *exogenous*:

$$\mathbb{E}[Xe] = 0 \quad \text{or} \quad \mathbb{E}[e|X] = 0 \text{ for all } X$$

While this works for many models, this assumption is not always satisfied.

When Can Endogeneity Fail?

1. Suppose we wanted to estimate the effect of an extra year of education on wages?

$$WAGE_i = \beta_0 + \beta_1 EDUC_i + e_i \tag{34}$$

Unobserved "ability" might increase education and earnings!

- More accomplished students are accepted to universities.
- More productive people earn a higher wage.

2. Suppose we wanted to estimate the effect of price on the demand of bottled water?

$$Q_i^D = \beta_0 + \beta_1 PRICE_i + e_i \tag{35}$$

A natural disaster might hit both!

- If tap water is shut off, people will demand more bottle water.
- If supply chains are disrupted, then firms might charge more.

Why does it matter?

In these cases, we have $\text{Cov}(X, e) \neq 0$, but what affect does this have on our OLS estimates?

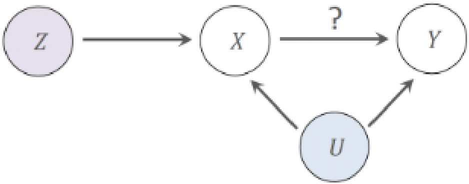
$$\begin{aligned}
\mathbb{E}[\hat{b}_1] &= \frac{\text{Cov}(X, Y)}{\text{Var}(X)} \\
&= \frac{\text{Cov}(X, \beta_0 + \beta_1 X + e)}{\text{Var}(X)} \\
&= \frac{\text{Cov}(X, \beta_0)}{\text{Var}(X)} + \frac{\text{Cov}(X, \beta_1 X)}{\text{Var}(X)} + \frac{\text{Cov}(X, e)}{\text{Var}(X)} \\
&= \beta_1 + \frac{\text{Cov}(X, e)}{\text{Var}(X)}
\end{aligned}$$

When X is endogenous, then our estimator is biased!

How do we solve it?

One solution to the problem of endogeneity are instruments, denoted as Z. They must satisfy the following conditions

- (Exclusion) Z is uncorrelated with the regression error e, i.e. $\text{Cov}(Z, e) = 0$
- (Relevance) Z is (strongly) correlated with X



How do we use an IV?

Suppose the model is given

$$Y_i = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_B X_{iB} + \theta_1 W_{i1} + \cdots + W_{iG} + e_i \quad (36)$$

where X is the endogenous variable and W is an exogeneous variables Suppose we have a good instrument for X. How do we use it?

Solution: 2SLS

One solution is the Two Stage Least Squares (2SLS):

1. Estimate the regression of X on Z and W:

$$X_i = \theta_0 + \beta_1 Z_i + \beta_2 W_i + v_i \tag{37}$$

   Store the fitted values $\hat{X}$

   - This results in $\text{Cov}(\hat{X}, e) = 0$

2. Estimate the regression of Y on $\hat{X}$ and W:

$$Y_i = \beta_0 + \beta_1 \hat{X}_i + \beta_2 W_i + e_i \tag{38}$$

Your estimates of the second stage will be consistent for β_1!

Solution: 2SLS

If there's no additional exogenous variables (W), the estimate for β_1 is

$$\tilde{b}_1 = \frac{\sum_{i=1}^{N}(z_i - \bar{z})(y_i - \bar{y})}{\sum_{i=1}^{N}(z_i - \bar{z})(x_i - \bar{x})} \tag{39}$$

The standard error of $\tilde{b}_1$ given by

$$\mathrm{Var}(\tilde{b}_1) = \frac{\sum(z_i - \bar{z})^2 \hat{\sigma}_{IV}^2}{\left[\sum(z_i - \bar{z})(x_i - \bar{x})\right]^2} \approx \frac{\mathrm{Var}(\hat{b}_1^{OLS})}{(Corr(X, Z))^2} \tag{40}$$

Solution: 2SLS

Some things to keep in mind:

1. You always have to include the exogenous variable W in the first stage.
2. Manually running both models and regressing the second stage on $\hat{X}$ is wrong.
   - $\longrightarrow$ Parameters are still correct (unbiased/consistent),
   - $\longrightarrow$ but standard errors are wrong (inefficient).
3. Want to use a command like

$$\texttt{lm <- ivreg(Y} \sim \texttt{X} + \texttt{W} \mid \texttt{Z} + \texttt{W}, \texttt{data} = \ldots)$$

Instrument Selection

Earlier, we said that

$$\text{Var}(\tilde{b}_1^{IV}) \approx \frac{\text{Var}(\hat{b}_1^{OLS})}{(Corr(X,Z))^2}$$

so we want to select instruments that are highly correlated with X. This keeps our standard errors as small as possible!

- With only 1 instrument, a first-stage F-stat greater than 10 is a good rule of thumb
- With more instruments, you want an even higher F-stat, but no rule of thumb exists.

Checking for Endogeneity (Hausman Test)

To check whether X and e are correlated, conduct a Hausman Test

1. Estimate your first stage regression and store the residuals:

$$\hat{v}_i = X_i - (\hat{\theta}_0 + \hat{\theta}_1 Z_i + \hat{\theta}_2 W_i)$$

2. Include the residuals in the main regression:

$$Y_i = \beta_0 + \beta_1 X_i + \delta \hat{v}_i + e_i$$

3. Conduct the following test:

$$\begin{aligned} H_0 &: \delta = 0 \quad \text{(No endogeneity)} \\ H_1 &: \delta \neq 0 \quad \text{(Endogeneity)} \end{aligned}$$

How many IVs do we need?

Denote L as the amount of instruments we have:

- $L < B$: The model is *underidentified* and the model can't be estimated.
- $L = B$: The model is *just-identified* and the parameter estimates are unique
- $L > B$: The model is *overidentified*, and problems may arise as your first stage estimate $\hat{X}$ approach the original X values in the sample (generally ok).

Overidentification (Sargan) Tests

Only when $L > B$, you can conduct a Sargan test. This tells you whether you additional instruments should be included.

1. Conduct the 2SLS estimation, and store the residuals of the second stage:

$$\hat{e} = \hat{Y} - (\hat{\beta}_0 + \hat{\beta}_1 X + \hat{\beta}_2 W) \tag{41}$$

2. Regress $\hat{e}$ on all available instruments:

$$\hat{e} = \theta_0 + \theta_1 Z_1 + \cdots + \theta_L Z_L \tag{42}$$

3. If the instruments are all valid, then

$$N \times R^2 \sim \chi^2_{(L-B)}$$

4. If the test statistic is larger than the critical value, then at least one surplus instrument is not valid.

Simultaneous Equations

Simultaneous Equations are used in cases where we believe equilibria occur. Models of supply and demand are a popular case. In general, these models describe systems where multiple endogenous variables are determined simultaneously by a set of equations.

Big Topics:

1. Identification
2. Reduced Form
3. Estimation by 2SLS

Identification

Whenever we have a simultaneous equation system, we always break down the variables into endogenous variables and exogenous variables:

- M Endogenous variables and equations
- K Exogenous variables

For identification (i.e. good estimation), we need each of the M equations to exclude $M - 1$ exogenous variables.

Reduced Form Equations

One way to describe simultaneous equation systems is by solving for the reduced form equations.

The reduced form system yields a system of M equations, that each contain *one* endogenous variable on the left-hand, and *all* the exogenous variables on the right hand side.

When solving for these, treat it as a system of equations where the endogenous variables are the "unknown" variables that you are solving for.

Reduced Form Equations

Structural Equations:

$$y_{i1} = \alpha_1 + \alpha_2 y_{i2} + \alpha_3 x_{i1} + e_{i1}$$
$$y_{i2} = \beta_1 + \beta_2 y_{i1} + \beta_3 x_{i2} + e_{i2}$$

Reduced Form Equations

Structural Equations:

$$y_{i1} = \alpha_1 + \alpha_2 y_{i2} + \alpha_3 x_{i1} + e_{i1}$$
$$y_{i2} = \beta_1 + \beta_2 y_{i1} + \beta_3 x_{i2} + e_{i2}$$

Reduced Form Equations:

$$y_{i1} = \pi_{11} + \pi_{12} x_{i1} + \pi_{13} x_{i2} + v_{i1}$$
$$y_{i2} = \pi_{21} + \pi_{22} x_{i1} + \pi_{23} x_{i2} + v_{i2}$$

Estimation by 2SLS

Estimation of a simultaneous equation system relies on the reduced form equation:

1. Estimate the reduced form equation for the endogenous variable(s) on the right hand side of the equation. Treat these as the first stage.
2. Run a regression of the original equation, replacing the true values of the RHS endogenous variables with their reduced form fitted values

If you don't do this, then estimates will be biased and inconsistent!